Related-Work Literature Review for FinReg-SLM

A Domain-Specialized Small Language Model for Indian Financial & Regulatory Compliance

(Prepared as a lift-in-ready Related Work chapter. Peer-reviewed venues prioritized; arXiv-only works quarantined to a labelled appendix per the project constraint. Where a venue or number could not be verified, this is stated explicitly rather than guessed.)

TL;DR

- **A strong, version-aware RAG baseline — not fine-tuning — should be the load-bearing component of FinReg-SLM.** The best peer-reviewed head-to-head (Ovadia et al., EMNLP 2024) shows retrieval-augmented generation consistently beats unsupervised fine-tuning for injecting *both* seen and entirely new factual knowledge; fine-tuning must therefore be justified for *style, format, citation behaviour and refusal calibration*, not for storing regulatory facts that change constantly. (ACL Anthology)
- **The specific gap FinReg-SLM targets is real and defensible: there is no peer-reviewed NLP work that processes RBI circulars, SEBI regulations, Indian GST/CBIC circulars, the Income-tax Acts, or the DPDP Act for compliance question-answering.** Peer-reviewed Indian legal NLP is overwhelmingly *court-judgment*-centric (InLegalBERT, IL-TUR, NyayaAnumana); financial-regulatory NLP that is peer-reviewed is US/EU-centric (SEC, Fed, ESMA, GDPR). The Indian-regulatory works that do exist are 2025–2026 arXiv preprints only.
- **The literature issues three concrete warnings we must design around and report honestly:** (1) RAG does *not* eliminate hallucination — leading commercial legal-RAG tools still hallucinate 17–33% of the time (Magesh et al. 2025); (2) domain fine-tuning causes catastrophic forgetting that *worsens with model scale* (Luo et al. 2023); (3) parameter-efficient fine-tuning trades capability for stability — LoRA "learns less and forgets less" than full fine-tuning (Biderman et al., TMLR 2024).

Key Findings (orientation for the reader)

1. **Domain specialization is well-established but the payoff is task-dependent.** Domain-adaptive pretraining (Gururangan et al., ACL 2020) and biomedical models (BioBERT, PubMedBERT) reliably beat general models on *classification/extraction*; the evidence that a small fine-tuned model beats a *frontier* model on open-ended domain QA is much thinner, and IL-TUR even shows GPT-4 beating specialized SOTA on some Indian legal metrics. Our novelty claim must be scoped to a *retrieval-grounded, version-aware, citation-checked* setting, not "small model beats GPT."
2. **PEFT (LoRA/QLoRA) makes the engineering feasible** on a student budget and, helpfully, is the *forgetting-safest* way to adapt.

3. **RAG-vs-fine-tuning is the central methodological risk**, and the peer-reviewed verdict favours RAG for knowledge — reframing our fine-tuning contribution.
4. **Hallucination/citation-grounding and catastrophic-forgetting measurement** are mature enough that we can adopt standard methodology; failing to report them would be a reviewer red flag.
5. **Temporal/versioned knowledge** is an active but still-immature research area — and it is exactly where the Indian setting (Income-tax Act 2025 renumbering, RBI Master Direction consolidation, DPDP phased commencement) gives us a distinctive contribution.

Details — Thematic Sections

Section 1 — Domain-specialized and domain-adapted language models

Gururangan, Marasović, Swayamdipta, Lo, Beltagy, Downey & Smith. "Don't Stop Pretraining: Adapt Language Models to Domains and Tasks." ACL 2020, pp. 8342–8360. Anthology 2020.acl-main.740; DOI 10.18653/v1/2020.acl-main.740. *(Honorable mention, best paper.)*

- **Contribution/method:** A second phase of self-supervised pretraining on in-domain text (domain-adaptive pretraining, DAPT) and on task text (task-adaptive pretraining, TAPT), studied across four domains (biomedical, CS, news, reviews) and eight classification tasks.
- **Results:** Consistent gains from DAPT and TAPT under both high- and low-resource settings; simple data-selection augmentation is an effective cheap alternative.
- **Limitations:** Confined to ~RoBERTa-scale encoders and *classification* tasks; predates generative LLMs and RAG.
- **Relevance to FinReg-SLM:** The foundational justification that adapting to a narrow corpus (Indian regulation) helps. But it warns us: the demonstrated gains are for classification, not for open-ended, citation-bearing QA — so we should not over-claim that DAPT-style adaptation alone will fix factual QA. It also motivates a cheap intermediate: continued pretraining on the regulatory corpus before any instruction tuning.

Lee, Yoon, Kim, Kim, Kim, So & Kang. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining." ***Bioinformatics*** **36(4):1234–1240, 2020. DOI 10.1093/bioinformatics/btz682.**

- **Contribution/method:** Continual pretraining of BERT on PubMed abstracts + PMC full text (same vocabulary as BERT).
- **Results:** Improves biomedical NER (+0.62 F1), RE (+2.80 F1) and QA (+12.24 MRR) over the prior SOTA.
- **Limitations:** Encoder-only; reuses general vocabulary (suboptimal tokenization for domain terms); gains do not scale like general benchmarks.
- **Relevance to FinReg-SLM:** The canonical "continual-pretrain from a general checkpoint" recipe — cheaper than from-scratch and directly analogous to adapting Llama 3.1 8B / Qwen

2.5 7B on Indian regulatory text. The tokenizer caveat is a live concern for our corpus (statutory section numbers, "Section 194-IA", GSTIN patterns, ₹ amounts).

Gu, Tinn, Cheng, Lucas, Usuyama, Liu, Naumann, Gao & Poon. "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing" (PubMedBERT). ***ACM Transactions on Computing for Healthcare*** **3(1), 2022. DOI 10.1145/3458754.**

- **Contribution/results:** Pretraining *from scratch* on in-domain text with an in-domain vocabulary beats mixed-domain continual pretraining across most BLURB tasks, often by a wide margin; RoBERTa (largest general corpus) is among the worst on biomedical tasks.
- **Relevance to FinReg-SLM:** Counter-intuitively argues that *more general data ≠ better domain performance* — supportive of specialization. But "from scratch" is infeasible at 7–8B on our budget, so we should lean on the BioBERT-style continual-pretrain path and treat PubMedBERT as evidence that in-domain vocabulary/tokenization matters.

Finance models. *FinQA* (Chen, Chen, Smiley, Shah, et al., **EMNLP 2021**, pp. 3697–3711, Anthology 2021.emnlp-main.300; DOI 10.18653/v1/2021.emnlp-main.300) is a numerical-reasoning-over-financial-data QA dataset — peer-reviewed and directly relevant as a *methodological template* for building numeric compliance questions (e.g., TDS/GST computation). *PIXIU / FinMA* (Xie, Han, Zhang, Lai, Peng, Lopez-Lira & Huang, **NeurIPS 2023 Datasets & Benchmarks Track**, pp. 33469–33484; OpenReview vTrRq6vCQH) contributes an open financial LLM plus the FLARE evaluation suite — the clearest peer-reviewed precedent for an open, instruction-tuned domain financial model with a companion benchmark, which is structurally what we propose. *(Note: the NeurIPS camera-ready title is "PIXIU: A Comprehensive Benchmark, Instruction Dataset and Large Language Model for Finance" — cite this, not the differently-worded preprint title.)*

- **Counter-evidence / honesty flags:** *FinBERT* (Araci, 2019) and *BloombergGPT* (Wu et al., 2023) are frequently cited but are, respectively, an **arXiv preprint** and a **proprietary technical report** — neither is peer-reviewed, and BloombergGPT is closed and irreproducible. We should cite them only as context, and we note that BloombergGPT's reported edge over general models was modest and five-shot only.
- **Relevance to FinReg-SLM:** FinQA and PIXIU/FinMA justify the "open small domain model + purpose-built benchmark" formula in peer-reviewed form. They also warn that finance-LLM gains are uneven and that reproducibility/openness is itself a contribution — favouring our open-weight, documented approach.

Section 2 — Small language models, PEFT, and the cost/latency argument

Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang & Chen. "LoRA: Low-Rank Adaptation of Large Language Models." ICLR 2022.

- **Method/results:** Freeze pretrained weights; inject trainable low-rank matrices into attention projections, cutting trainable parameters by orders of magnitude with negligible inference

latency (adapters merge into weights).

- **Relevance:** The default adaptation mechanism for FinReg-SLM; near-zero added latency is important for our cost/latency comparison against frontier APIs.

Dettmers, Pagnoni, Holtzman & Zettlemoyer. "QLoRA: Efficient Finetuning of Quantized LLMs." NeurIPS 2023.

- **Method/results:** Back-propagate through a frozen 4-bit (NF4) quantized base model into LoRA adapters. Per the paper, QLoRA is "an efficient finetuning approach that reduces memory usage enough to **finetune a 65B parameter model on a single 48 GB GPU while preserving full 16-bit finetuning task performance**" (reducing the requirement from >780 GB to 48 GB).
- **Relevance to FinReg-SLM:** This is the enabling result that makes a student major project viable — a 7–8B model is comfortably fine-tunable on a single consumer/college GPU, and even much larger models are within reach, so compute is not a limiting excuse for our experimental design.

Houlsby et al., "Parameter-Efficient Transfer Learning for NLP" (Adapters), ICML 2019; Li & Liang, "Prefix-Tuning", ACL 2021. Establish the broader PEFT family; useful as ablation baselines against LoRA/QLoRA.

- **Honest caveat on the cost/latency/energy claim:** I did **not** locate a single definitive peer-reviewed study that cleanly benchmarks small-vs-frontier deployment cost/latency/energy for a compliance-style workload; such numbers are scattered and often vendor-sourced. We should therefore *generate our own* cost-per-query and latency measurements (a genuine contribution) rather than cite a headline figure we cannot verify.

Section 3 — Retrieval-augmented generation vs fine-tuning (the central methodological risk)

Lewis, Perez, Piktus, Petroni, Karpukhin, Goyal, Küttler, Lewis, Yih, Rocktäschel, Riedel & Kiela. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." NeurIPS 2020, pp. 9459–9474.

- **Contribution:** Combines a parametric generator with a non-parametric retrieved index; improves knowledge-intensive tasks and supports provenance and cheap knowledge updates.
- **Relevance:** The architectural blueprint for our citation-grounded, updateable system; its emphasis on provenance is exactly what regulatory answers require.

Ovadia, Brief, Mishaeli & Elisha. "Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs." EMNLP 2024, pp. 237–250. Anthology 2024.emnlp-main.15.

- **Method:** Compares unsupervised fine-tuning vs RAG on knowledge-intensive tasks (MMLU subsets + a "current events" set for genuinely new knowledge), on Mistral-7B and others.

- **Results (verbatim framing):** "while unsupervised fine-tuning offers some improvement, **RAG consistently outperforms it, both for existing knowledge encountered during training and entirely new knowledge … LLMs struggle to learn new factual information through unsupervised fine-tuning**." On the current-events task RAG reached **0.875 (Mistral)** vs **0.504** for regular fine-tuning; exposing the model to many paraphrases of a fact helps fine-tuning but still trails RAG.
- **Limitations:** Uses *unsupervised* fine-tuning (not supervised instruction tuning), and MMLU-style MCQ; does not test citation faithfulness.
- **Relevance to FinReg-SLM — decisive:** This is the strongest peer-reviewed answer to "does a good RAG baseline make fine-tuning unnecessary *for facts*?" — yes, largely. Therefore FinReg-SLM must (a) build a genuinely strong RAG baseline first, and (b) reposition fine-tuning as improving *citation formatting, Indian-regulatory register, abstention, and answer structure* — knowledge that supervised instruction tuning (a setting Ovadia did not fully cover) can plausibly still add. This reframing is what protects our novelty from the "why not just RAG?" reviewer.

Section 4 — Legal & regulatory NLP, especially Indian

Paul, Mandal, Goyal & Ghosh. "Pre-trained Language Models for the Legal Domain: A Case Study on Indian Law" (InLegalBERT / InCaseLawBERT). ICAIL 2023.

- **Method/results:** Continual pretraining of LegalBERT variants on ~5.4M Indian court cases; InLegalBERT beats LegalBERT and baselines on legal statute identification, rhetorical-role/semantic segmentation, and court-judgment prediction (e.g., ECtHR-B macro-F1 75.88 vs 74.7). ~110M params; >1.8M HuggingFace downloads.
- **Relevance:** The strongest peer-reviewed proof that Indian-domain adaptation helps — but for *judgments*, not regulatory instruments, and at encoder scale. A key adjacent-precedent to cite and to distinguish ourselves from.

Joshi, Paul, Sharma, Goyal, Ghosh & Modi. "IL-TUR: Benchmark for Indian Legal Text Understanding and Reasoning." ACL 2024, pp. 11460–11499. Anthology 2024.acl-long.618.

- **Contribution:** Eight monolingual (English/Hindi) and multilingual Indian legal tasks with a public leaderboard.
- **Results/nuance:** Specialized SOTA generally beats GPT-3.5/GPT-4 across most tasks, **but GPT-4 exceeds SOTA on legal machine-translation on BLEU/GLEU** — a documented counter-example to the "specialized always wins" narrative.
- **Relevance:** The methodological template for a multi-task Indian benchmark and the honest reporting we should emulate; its GPT-4 exception is exactly the kind of counter-evidence our evaluation must be prepared to find and report.

Nigam, Balaramamahanthi, Mishra, Shallum, Ghosh & Bhattacharya. "NyayaAnumana & INLegalLlama: The Largest Indian Legal Judgment Prediction Dataset and Specialized

Language Model for Enhanced Decision Analysis." COLING 2025, pp. 11135–11160. Anthology 2025.coling-main.738.

- **Method/results:** A **702,945-case** corpus; a two-phase recipe (continual pretraining for domain-knowledge injection, then supervised fine-tuning). Per the paper, "incorporating diverse court data significantly boosts model accuracy, **achieving approximately 90% F1-score in prediction tasks**"; the generative INLegalLlama reaches **76.05% accuracy**, above reported PredEx (51.77%) and base Llama-2 (57.26%). Semantic Scholar
- **Relevance to FinReg-SLM — direct:** This is the closest peer-reviewed methodological cousin: an Indian, open, two-phase (continual-pretrain → SFT) domain LLM with a purpose-built dataset. We can borrow its pipeline design and explainability emphasis wholesale. It also confirms the *domain* gap: it targets judgment prediction, not regulatory compliance QA.

International regulatory / evaluation resources (peer-reviewed):

- **Guha, Nyarko, Ho, Ré, Chilton, et al. "LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in LLMs." NeurIPS 2023 Datasets & Benchmarks Track** (162 expert-crafted tasks over six reasoning types). Template for expert-authored task taxonomy; note their explicit contamination caveat (some tasks may have leaked into pretraining) — a lesson for our benchmark.
- **Hendrycks, Burns, Chen & Ball. "CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review." NeurIPS 2021 Datasets & Benchmarks Track** (510 contracts, 41 clause types, 13,101 expert annotations). The gold standard for expert-annotated legal extraction methodology and annotation scale.
- **Holzenberger, Blair-Stanek & Van Durme. "A Dataset for Statutory Reasoning in Tax Law Entailment and Question Answering" (SARA). NLLP@KDD 2020**; refined in **Holzenberger & Van Durme, "Factoring Statutory Reasoning as Language Understanding Challenges," ACL 2021, pp. 2742–2758**; later a LegalBench component. The most directly relevant *statutory/tax* reasoning benchmark (US Internal Revenue Code) — a design model for our Income-tax computation questions and the neuro-symbolic caution that pure LLMs do statutory arithmetic poorly.
- **The gap (subagent-verified):** No peer-reviewed, non-arXiv NLP paper specifically processes **RBI circulars/Master Directions, SEBI regulations, Indian GST/CBIC circulars, the Income-tax Acts, or the DPDP Act** for compliance/QA/classification. Peer-reviewed financial-regulatory NLP is US/EU-centric (e.g., the COLING 2025 FinNLP "Regulations Challenge" over SEC/Fed/FDIC/ESMA/XBRL; GDPR compliance-checking in *IEEE TSE*). Indian regulatory NLP presently exists only as 2025–2026 arXiv preprints and non-academic vendor white-papers. **This is the crux of FinReg-SLM's contribution.**

Section 5 — Hallucination, factuality and citation grounding

Dahl, Magesh, Suzgun & Ho. "Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models." *Journal of Legal Analysis* 16(1):64–93, 2024. DOI 10.1093/jla/laae003.

- **Method/results:** First systematic profiling of legal hallucination in public LLMs on verifiable US federal-case questions at temperature 0. Per the paper, "legal hallucinations are alarmingly prevalent, **occurring between 58% of the time with ChatGPT-4 and 88% with Llama 2**" (GPT-3.5 ≈69%), models often fail to correct users' false legal premises, and are poorly calibrated about their own hallucinations.
- **Relevance:** Quantifies the danger our project addresses and supplies a hallucination *typology* and a temperature-0 protocol we should reuse. It motivates strict citation-grounding and abstention.

Magesh, Surani, Dahl, Suzgun, Manning & Ho. "Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools." *Journal of Empirical Legal Studies* 22(2):216–242, 2025. DOI 10.1111/jels.12413.

- **Method/results:** First preregistered evaluation of *commercial RAG* legal tools. Per the paper, "the AI research tools made by LexisNexis (Lexis+ AI) and Thomson Reuters (Westlaw AI-Assisted Research and Ask Practical Law AI) **each hallucinate between 17% and 33% of the time**" — approximately Lexis+ AI ~17%, Westlaw AI-Assisted Research ~33%, versus GPT-4 at ~43% — refuting vendor "hallucination-free" claims.
- **Relevance to FinReg-SLM — decisive caution:** This is the single most important warning for our design: **RAG reduces but does not eliminate hallucination.** Our evaluation must measure grounded-citation accuracy and mis-citation/mis-grounding explicitly, define a hallucination typology for regulatory answers (fabricated section, wrong version, mischaracterized provision), and treat a non-zero residual hallucination rate as an expected, reportable outcome rather than a failure of the project.

(Related peer-reviewed factuality tooling worth citing: Min et al., "FActScore," EMNLP 2023 — fine-grained atomic factual-precision evaluation for long-form generation; useful for scoring our long compliance answers.)

Section 6 — Catastrophic forgetting in domain fine-tuning, and mitigation

Luo, Yang, Meng, Li, Zhou & Zhang. "An Empirical Study of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning." EMNLP 2023 *(pages reported as 8934–8947 in downstream citations; several later works cite it as CoRR/arXiv 2308.08747 — cite the EMNLP proceedings version and, if final page numbers cannot be confirmed against the anthology, flag them as to-verify).*

- **Method/results:** Measures forgetting of domain knowledge, reasoning, and reading comprehension during continual instruction tuning across 1B–7B models. Catastrophic forgetting is generally observed and **its severity intensifies as model scale increases**; general instruction tuning beforehand can partially alleviate later forgetting.
- **Relevance:** Supplies the standard *methodology* for the "general-capability regression" our task explicitly requires us to report — measure a held-out general benchmark (e.g.,

MMLU/commonsense/reasoning) before vs after fine-tuning, at multiple checkpoints. The scale finding tells us our 7–8B target is squarely in the risk zone.

Biderman, Portes, Gonzalez Ortiz, Paul, Greengard, Jennings, King, Havens, Chiley, Frankle, Blakeney & Cunningham. "LoRA Learns Less and Forgets Less." *Transactions on Machine Learning Research* (TMLR), 2024 (Featured Certification).

- **Method/results:** On code and math target domains, LoRA generally underperforms full fine-tuning on the *target* task ("learns less") but **better preserves out-of-domain base capability ("forgets less")** — and does so more effectively than classical regularizers like weight decay/dropout, while maintaining more output diversity.
- **Relevance to FinReg-SLM — direct:** Justifies our choice of LoRA/QLoRA on *forgetting-safety* grounds, and sets the honest expectation of a *capability/forgetting trade-off* we should quantify. It gives us a principled ablation: full FT vs LoRA vs QLoRA on both the compliance benchmark and a general-capability suite. *(Cite as TMLR 2024; a separate COLM 2024 presentation could not be confirmed.)*

(Foundational anchor: McCloskey & Cohen, "Catastrophic interference in connectionist networks," Psychology of Learning and Motivation, 1989 — the origin of the phenomenon; EWC (Kirkpatrick et al., PNAS 2017) is the canonical mitigation reference if we discuss regularization-based methods.)

Section 7 — Temporal reasoning, knowledge staleness, versioned corpora

Chen, Wang & Wang. "A Dataset for Answering Time-Sensitive Questions" (TimeQA). NeurIPS 2021 Datasets & Benchmarks Track.

- **Method/results:** Time-sensitive questions over time-evolving Wikidata facts aligned to Wikipedia, split into Easy (explicit time expressions) and Hard (implicit, requiring temporal reasoning), ~20k QA pairs per split. Fine-tuned models reached only ~33% (Easy) / ~27% (Hard) — far below their ~60%+ on NQ/TriviaQA — exposing weak temporal reasoning.
- **Relevance to FinReg-SLM — high:** The canonical peer-reviewed demonstration that models handle *point-in-time* facts poorly. This is precisely FinReg-SLM's differentiator: the **Income-tax Act 2025 replaced the 1961 Act from 1 April 2026 with full section renumbering; RBI consolidated many circulars into Master Directions; DPDP Rules 2025 have phased commencement dates.** TimeQA's Easy/Hard split design directly inspires an "as-of-date" axis in our benchmark (e.g., "As of FY 2024-25 vs FY 2026-27, what is the applicable provision?"), and it warns that answers must be *version-tagged* or they will be silently wrong.
- **Honesty note:** Related peer-reviewed temporal resources (StreamingQA, SituatedQA, TempReason, TimeBench) exist and are relevant, but I could not individually re-verify each venue within budget; TimeQA is the one fully confirmed anchor (NeurIPS D&B 2021). Knowledge-editing methods (ROME/MEMIT-style) are relevant to "updating" but I did not verify their venues here and flag them as to-check rather than cite loosely.

Section 8 — Benchmark and evaluation-set construction methodology

Zheng, Chiang, Sheng, Zhuang, Wu, Zhuang, Lin, Li, Li, Xing, Zhang, Gonzalez & Stoica. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." NeurIPS 2023 Datasets & Benchmarks Track.

- **Method/results:** Establishes GPT-4-as-judge as a scalable proxy (~85% agreement with humans, exceeding ~81% human–human agreement) *and* rigorously documents its biases: **position bias, verbosity bias, self-enhancement bias, and limited reasoning/math judging**, with mitigations.
- **Relevance to FinReg-SLM — direct:** If we use LLM-as-judge to scale evaluation, we must (a) validate against expert labels on a subset, (b) counter position/verbosity bias (randomize order, control length), and (c) avoid self-enhancement bias (don't let a model family judge itself). Citing this pre-empts the "your automated eval is unreliable" critique.

Gebru, Morgenstern, Vecchione, Vaughan, Wallach, Daumé III & Crawford. "Datasheets for Datasets." *Communications of the ACM* 64(12):86–92, 2021. DOI 10.1145/3458723.

- **Contribution:** A 57-question / 7-section standard (motivation, composition, collection, preprocessing, uses, distribution, maintenance) for documenting datasets.
- **Relevance:** Since we are *releasing a new hand-verified Indian compliance benchmark*, shipping a datasheet is the defensible documentation standard and directly strengthens the paper's contribution and reproducibility. Pair with LegalBench/CUAD as evidence for **expert-annotation protocols and inter-annotator-agreement reporting**, and with Zheng et al. on judge validity.
- **Contamination:** LegalBench's own contamination caveat (Section 4) is the peer-reviewed hook for a contamination-control subsection; we should build questions from *post-cutoff* instruments (Income-tax Act 2025, DPDP Rules 2025, Nov-2025 RBI Master Directions) precisely because they are less likely to be in any model's pretraining.

Section 9 — Multilingual / Indic NLP relevant to Indian regulatory text (brief)

Kakwani, Kunchukuttan, Golla, Gokul N.C., Bhattacharyya, Khapra & Kumar. "IndicNLPSuite: Monolingual Corpora, Evaluation Benchmarks and Pre-trained Multilingual Language Models for Indian Languages." Findings of EMNLP 2020, pp. 4948–4961. Anthology 2020.findings-emnlp.445; DOI 10.18653/v1/2020.findings-emnlp.445. Introduces IndicCorp, the IndicGLUE benchmark, and IndicBERT — the peer-reviewed backbone for Indic evaluation and tokenizer/fertility considerations.

Ramesh, Doddapaneni, Bheemaraj, et al. "Samanantar: The Largest Publicly Available Parallel Corpora Collection for 11 Indic Languages." *TACL* 10:145–162, 2022. DOI 10.1162/tacl_a_00452. Peer-reviewed AI4Bharat resource; relevant if we extend to Hindi/regional-language regulatory text. *(IndicNLG Benchmark, EMNLP 2022, pp. 5363–5394, Anthology 2022.emnlp-main.360, is an alternative peer-reviewed AI4Bharat anchor for generation tasks.)*

- **Relevance/scope note:** Indian primary regulatory sources (RBI/SEBI/CBIC/tax/DPDP) are authoritative **in English**, so multilinguality is *not* core to FinReg-SLM's accuracy. Its value is **accessibility** — serving Hindi/regional-language queries against English-authoritative text — and **tokenizer fertility** awareness if we localize. Keep this section short and framed as future work, not a central claim.

(a) Synthesis — The precise, defensible research gap and how to phrase novelty

What has been done (peer-reviewed): (i) Domain adaptation reliably helps on classification/extraction (Gururangan; BioBERT; PubMedBERT). (ii) Indian legal NLP is mature but **judgment-centric** (InLegalBERT/ICAIL 2023; IL-TUR/ACL 2024; NyayaAnumana/COLING 2025). (iii) Statutory/tax reasoning and contract review have expert benchmarks — but **US/EU** (SARA; CUAD; LegalBench). (iv) The RAG-vs-fine-tuning question has a peer-reviewed answer favouring RAG for knowledge (Ovadia/EMNLP 2024). (v) Hallucination in legal RAG is quantified (Dahl 2024; Magesh 2025). (vi) Forgetting and PEFT-forgetting trade-offs are characterized (Luo 2023; Biderman/TMLR 2024). (vii) Temporal QA is a recognized hard problem (TimeQA/NeurIPS 2021). (viii) Benchmark/judge/documentation methodology is standardized (LegalBench; CUAD; Zheng 2023; Gebru 2021).

What has *not* been done: There is **no peer-reviewed system or benchmark for Indian financial-regulatory compliance QA** grounded in RBI/SEBI/CBIC-GST/Income-tax/DPDP primary sources, and **no peer-reviewed treatment of the version/temporal problem for Indian regulation** (the 1961→2025 Income-tax renumbering, RBI Master Direction consolidation, DPDP phased commencement). Existing Indian-regulatory efforts are arXiv-preprint-only.

How to phrase the novelty claim so it survives review (recommended wording):

> "We present the first *peer-reviewed*, version-aware, citation-grounded compliance-QA system and expert-verified benchmark for **Indian** financial and regulatory primary sources (RBI, SEBI, CBIC-GST, Income-tax Acts 1961 & 2025, DPDP Act & Rules), and the first controlled study of RAG vs LoRA/QLoRA fine-tuning under **temporal versioning** in this domain, reporting general-capability regression honestly."

Do **not** claim "small model beats frontier LLM" as the headline (IL-TUR shows GPT-4 can win on some metrics; Ovadia shows RAG dominates fine-tuning for facts). Claim instead: **jurisdiction + temporal-versioning + citation-grounding + a new expert benchmark + an honest cost/latency/forgetting accounting.** That combination is genuinely unoccupied and each element is individually defensible.

Honesty check on gap width: The gap is narrower than "nobody has done Indian regulatory NLP" — court-judgment NLP is crowded, and 2025–2026 preprints (IndiaFinBench, ComplianceNLP) are staking nearby claims. Our durable differentiators are the **temporal/version axis** and **citation-grounded abstention**, which even those preprints do not fully address.

(b) What the literature warns us about — top documented failure modes

1. **RAG will not eliminate hallucination.** Leading commercial legal-RAG tools still hallucinate 17–33% (Magesh et al. 2025). → We must measure and report a residual hallucination/mis-grounding rate, not claim to have solved it.
2. **Models uncritically accept false premises and are poorly calibrated about their own errors** (Dahl et al. 2024, 58–88% on verifiable questions). → Build contra-factual/false-premise questions and abstention tests into the benchmark.
3. **Fine-tuning struggles to inject new facts; RAG wins** (Ovadia et al., EMNLP 2024). → Do not rely on fine-tuning to memorize circulars; keep facts in the retrievable, version-tagged corpus.
4. **Catastrophic forgetting worsens with scale during continual/instruction fine-tuning** (Luo et al. 2023). → Measure general-capability regression at multiple checkpoints; prefer instruction-tuning-then-domain order.
5. **PEFT is a capability/stability trade-off — LoRA learns less even as it forgets less** (Biderman et al., TMLR 2024). → Expect and quantify a small target-task penalty for LoRA; ablate against full fine-tuning.
6. **Temporal/point-in-time reasoning is a known model weakness** (TimeQA; Chen et al. 2021, ~27–33% accuracy). → Version-tag every corpus chunk and every gold answer; without it, answers to "as-of-date" questions will be silently wrong.
7. **Benchmark contamination and stale gold answers** (LegalBench caveat; general temporal-staleness). → Prefer post-cutoff instruments (Income-tax Act 2025, DPDP Rules 2025) and document a contamination-control procedure.
8. **LLM-as-judge bias (position/verbosity/self-enhancement)** (Zheng et al. 2023). → Validate the judge against expert labels; randomize order, control length, and never let a model judge its own family.
9. **Statutory arithmetic is unreliable in pure LLMs** (SARA line of work; Holzenberger et al.). → For GST/TDS/tax computation questions, consider tool/symbolic assistance and report exact-match separately from retrieval accuracy.

(c) Suggested citation set for the paper's Related Work (ranked by importance)

Tier 1 — must cite (core to the argument):

1. Ovadia et al., *Fine-Tuning or Retrieval?*, EMNLP 2024 — RAG vs FT verdict.
2. Lewis et al., *RAG*, NeurIPS 2020 — architecture.
3. Magesh et al., *Hallucination-Free?*, J. Empirical Legal Studies 2025 — RAG still hallucinates.
4. Dahl et al., *Large Legal Fictions*, J. Legal Analysis 2024 — legal hallucination baseline.
5. Luo et al., *Catastrophic Forgetting in LLMs During Continual Fine-tuning*, EMNLP 2023.
6. Biderman et al., *LoRA Learns Less and Forgets Less*, TMLR 2024.
7. Hu et al., *LoRA*, ICLR 2022; Dettmers et al., *QLoRA*, NeurIPS 2023.

8. Gururangan et al., *Don't Stop Pretraining*, ACL 2020.
9. Guha et al., *LegalBench*, NeurIPS D&B 2023.
10. Chen et al., *TimeQA*, NeurIPS D&B 2021.
11. Gebru et al., *Datasheets for Datasets*, CACM 2021.
12. Zheng et al., *LLM-as-a-Judge / MT-Bench*, NeurIPS D&B 2023.

Tier 2 — strongly recommended (domain grounding): 13. Paul et al., *InLegalBERT*, ICAIL 2023. 14. Joshi et al., *IL-TUR*, ACL 2024. 15. Nigam et al., *NyayaAnumana & INLegalLlama*, COLING 2025. 16. Hendrycks et al., *CUAD*, NeurIPS D&B 2021. 17. Holzenberger et al., *SARA*, NLLP@KDD 2020 (+ ACL 2021 factoring paper). 18. Lee et al., *BioBERT*, Bioinformatics 2020. 19. Gu et al., *PubMedBERT*, ACM THCH 2022. 20. Chen et al., *FinQA*, EMNLP 2021. 21. Xie et al., *PIXIU/FinMA*, NeurIPS D&B 2023. 22. Kakwani et al., *IndicNLPSuite/IndicGLUE*, Findings of EMNLP 2020.

Tier 3 — supporting/optional: 23. Ramesh et al., *Samanantar*, TACL 2022 (or IndicNLG Benchmark, EMNLP 2022). 24. Min et al., *FActScore*, EMNLP 2023 (long-form factuality scoring). 25. Houlsby et al., *Adapters*, ICML 2019; Li & Liang, *Prefix-Tuning*, ACL 2021. 26. McCloskey & Cohen 1989 + Kirkpatrick et al. (EWC), PNAS 2017 (forgetting foundations).

This yields ~26 verified must/should-cite works grouped by theme — sufficient for both the project report's Related Work chapter and a conference paper.

(d) Suggested submission venues for a paper from this project

- **NLLP (Natural Legal Language Processing) Workshop** — co-located with EMNLP; the single best-fit venue for legal/regulatory NLP; typical submission deadline late summer (Aug–Sep) for an EMNLP-time workshop.
- **FinNLP / FNP / -in-Finance workshops* (co-located with EMNLP/COLING/IJCAI/COLING's FinNLP-FNP-LLMFinLegal) — best fit for the *financial-regulatory* framing; the COLING 2025 "Regulations Challenge" shows active interest in regulatory text.
- **JURIX** (International Conference on Legal Knowledge and Information Systems) — European, AI-&-Law community; typically December, deadlines around September.
- **ICAIL** (International Conference on AI and Law) — biennial flagship AI-&-Law venue (where InLegalBERT appeared); more selective, longer lead time.
- **Springer *Artificial Intelligence and Law*** (journal) — for an extended, benchmark-centric version.
- **ACL/EMNLP/COLING Findings or main (Resources & Evaluation track)** — appropriate if the *benchmark* is the headline contribution; LREC-COLING is especially benchmark-friendly.
- **Indian venues** — ICON (International Conference on NLP, India) and IEEE-indexed Indian conferences (e.g., COMAD/CODS, or IEEE regional conferences) for earlier-stage dissemination.

Deadlines vary year to year; treat the above as typical windows and confirm each year's CFP. NLLP@EMNLP is the recommended primary target; a FinNLP workshop is the recommended secondary.

Appendix — Preprint-only works (excluded from the main review at the user's request, listed for awareness; all UNREFEREED)

- **FinBERT** (Araci, 2019, arXiv 1908.10063) — widely cited financial sentiment model; unrefereed. Matters as the origin of "FinBERT" naming (note several *distinct* FinBERTs exist; some later variants did appear in venues such as IJCAI/Contemporary Accounting Research — cite those if used).
- **BloombergGPT** (Wu et al., 2023) — proprietary 50B finance LLM technical report; not peer-reviewed and irreproducible; contextual only.
- **FinGPT family** — open financial-LLM effort; predominantly preprint/workshop-report status; verify venue before citing as refereed.
- **IndiaFinBench; ComplianceNLP; RTI-Bench** (2025–2026, arXiv only) — the closest works to FinReg-SLM (Indian regulatory/financial text). They matter enormously because they stake nearby claims and confirm the domain is "underserved," but each is **unrefereed** as of the search date; treat as related-work-in-progress and as evidence of the gap, not as peer-reviewed prior art.
- **Ovadia et al.** first appeared as arXiv 2312.05934 but **is now peer-reviewed (EMNLP 2024)** — cite the EMNLP version (done above).
- **Dahl et al.** and **Magesh et al.** first appeared as preprints but **are now peer-reviewed** (JLA 2024; JELS 2025) — cite journal versions (done above).

Verification caveats carried forward: (1) Luo et al. 2023 — cite the EMNLP 2023 proceedings; confirm final page numbers against the ACL Anthology before camera-ready. (2) Biderman et al. — cite TMLR 2024; a separate COLM 2024 listing could not be confirmed. (3) PIXIU/FinMA — use the NeurIPS camera-ready title, which differs from the preprint. (4) StreamingQA/SituatedQA/TempReason and knowledge-editing (ROME/MEMIT) are relevant to Section 7 but their venues were not individually re-verified here; confirm before citing.